

VRUSHALI A. SURVE

Assistant Professor | Mathematics | Ph.D. (Pursuing)

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PROFESSIONAL PROFILE

Mathematics faculty and researcher with 10+ years of teaching experience in diploma and engineering education. Currently pursuing a Ph.D. in Mathematics at Parul University. Research interests include inventory modelling, deteriorating/perishable products, sustainable supply chains, carbon mitigation, optimization, and mathematical modelling. Experienced in teaching, paper setting/examination work, academic coordination, research publication, and faculty development activities.

RESEARCH PROFILE

- 3 research papers in Scopus-indexed journals (2026).
- Ph.D. research in mathematical inventory modelling and sustainable supply-chain optimization.
- Research themes: deterioration, carbon tax/cap, learning effects, preservation/green technology, price-sensitive demand and optimization.

EDUCATION

Qualification	Institution	Result / Status
Ph.D. in Mathematics	Parul University	Pursuing
M.Sc. in Mathematics	C.V. Raman University (CVRU)	60.7% First Class
B.Sc.	M.S. University	45%
H.S.C.	GSEB	50% Second Class
S.S.C.	GSEB	88.29% Distinction

PROFESSIONAL EXPERIENCE

Lecturer / Faculty | Parul Institute of Engineering & Technology / Parul University • Aug 2016 – Present

- Teaching, subject coordination, supervision, paper setting and examination responsibilities.

Assistant Professor | Sigma Institute of Engineering • Jan 2015 – May 2016

- Teaching, supervision and paper setting.

Lecturer | RMS Polytechnic • Feb 2014 – Sep 2014

- Teaching, supervision and paper setting.

Lecturer | Butler Polytechnic • Sep 2012 – Feb 2014

- Teaching, supervision and paper setting.

SELECTED RESEARCH PUBLICATIONS — SCOPUS-INDEXED JOURNALS

- 1. Optimising fresh produce demand with expiration date and carbon mitigation strategies under learning effect and nonlinear holding cost: a quadratic modelling approach of time-price-freshness for profit maximisation**
International Journal of Operational Research, 56(2), 131–156 (2026).
- 2. Integrated Economic Order Quantity Modeling with Deterioration, Emission Control, Learning Effects, and Preservation Investments under Inflation**
Process Integration and Optimization for Sustainability (2026). DOI: 10.1007/s41660-026-00791-0
- 3. Optimizing Sustainable Inventory with Deterioration, Learning, and Carbon Tax Under Time-price Sensitive Quadratic Demand and Inflationary Environment**
Process Integration and Optimization for Sustainability (2026). DOI: 10.1007/s41660-026-00801-1

OTHER PUBLICATIONS

- Production of Bio-Diesel from Waste Cooking Oil — IJCRT, Vol. 8, Issue 7 (2020), ISSN 2320-2882.
- A Research On: Study Of Challenges And Their Solutions Face By Women Entrepreneurs — IJCRT, Vol. 8, Issue 7 (2020), ISSN 2320-2882.
- A Research Study Of Challenges And Problems Faced By Working Women — IJCRT, Vol. 9, Issue 3 (2021), ISSN 2320-2882.

ACADEMIC & EXAMINATION RESPONSIBILITIES

- GTU Engineering Mathematics I & II — Paper Setter / Internal / External Examiner
- GTU Applied Mathematics — Paper Setter / Internal / External Examiner
- Parul University Applied Mathematics — Paper Setter / Internal / External Examiner

FACULTY DEVELOPMENT, WORKSHOPS & COURSES

- Advance Pedagogy — NITTR Kolkata, June 2023
- Data Science for Engineers — 8-week course, July–September 2023
- Teaching and Learning Pedagogy — July 2021
- Creating Online Courses through Open Educational Resources — July 2020
- Scilab — January 2020
- Communication: Diversity Management at Multiculture Workplace — June 2018
- Pedagogical Skills: Assessment and Grading — August 2018
- Professional Development and Intellectual Honesty — September 2018
- Leadership: Creating Directions, Setting Goals and Defining Task — October 2018

TECHNICAL & PROFESSIONAL SKILLS

• Mathematical Modelling & Optimization	• Inventory / Supply Chain Modelling
• Research & Academic Writing	• Mathematics Teaching & Curriculum Delivery
• Examination / Paper Setting	• MS Office & Internet
• C / C++ Fundamentals	•

CORE STRENGTHS

Positive attitude • Confidence • Adaptability • Academic commitment • Research orientation • Teamwork • Willingness to learn

LANGUAGES

English | Gujarati | Hindi | Marathi

HOBBIES

Reading | Photography | Listening to Music

PERSONAL DETAILS

Date of Birth	7 September 1985
Gender	Female
Location	Vadodara, Gujarat, India

CAREER OBJECTIVE

To contribute to high-quality mathematics education and research by combining strong academic teaching, mathematical modelling, research publication, and student development to prepare learners for professional and competitive environments.